MC-KELLY TAMUNOTENA PEPPLE

Lecturer I | AI Engineer | Machine Learning Researcher | Computer Vision Specialist | COREN Registered Engineer

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Google Scholar | LinkedIn | GitHub | Portfolio

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EXECUTIVE PROFILE

Experienced Lecturer I, Artificial Intelligence Engineer, and Machine Learning Researcher with over ten years of experience in higher education, engineering education, technical training, and applied research. Currently pursuing a Ph.D. in Electronic and Computer Engineering at the University of Nigeria, Nsukka, with research focused on Artificial Intelligence, Machine Learning, Computer Vision, and Intelligent Systems.

Experienced in teaching undergraduate engineering and computing courses, supervising more than ten undergraduate research projects, mentoring students, developing curricula, conducting interdisciplinary research, and publishing in peer-reviewed journals. Demonstrated expertise in integrating Artificial Intelligence and digital technologies into engineering education while delivering face-to-face, hybrid, and online instruction.

An academic leader with experience as Head of Department and committee chairman, contributing to curriculum development, programme accreditation, examination quality assurance, academic administration, and institutional improvement. A COREN Registered Engineer committed to research excellence, innovation, student success, and the application of intelligent technologies to solve real-world engineering challenges.

CORE COMPETENCIES

- Higher Education Teaching
- Artificial Intelligence & Machine Learning
- Research & Scholarly Publishing
- Curriculum Development
- Academic Leadership
- Student Supervision
- Computer Vision
- Control Systems
- Programme Accreditation
- Educational Technology

EDUCATION

Ph.D. (In View), Electronic and Computer Engineering
University of Nigeria, Nsukka | 2021–Present
Research Area: Artificial Intelligence, Computer Vision, Deep Learning, Intelligent Systems

M.Sc., Systems and Control
Coventry University, United Kingdom | 2013–2014

B.Tech., Computer Engineering
Rivers State University, Port Harcourt, Nigeria | 2003–2008

PROFESSIONAL CERTIFICATIONS

- Registered Engineer (COREN)
  Council for the Regulation of Engineering in Nigeria
- Python Developer Certificate
  FreeCodeCamp
- Intermediate Machine Learning
  Kaggle

RESEARCH GRANTS, SCHOLARSHIPS & SPONSORED RESEARCH

Tertiary Education Trust Fund (TETFund) Academic Staff Training and Development (AST&D) Scholarship
Sponsor: Tertiary Education Trust Fund (TETFund), Federal Government of Nigeria
Award Period: 2021 – Present
Recipient of the competitive Academic Staff Training and Development (AST&D) Scholarship supporting doctoral studies in Electronic and Computer Engineering at the University of Nigeria, Nsukka. The award supports advanced research in Artificial Intelligence, Machine Learning, Computer Vision, and Intelligent Systems.

ACADEMIC APPOINTMENT

Lecturer I
Department of Electrical/Electronic Engineering Technology
Federal Polytechnic of Oil and Gas, Bonny, Rivers State, Nigeria
2018 – Present
Responsibilities

- Teach undergraduate courses in Electrical/Electronic Engineering, Computer Engineering, Artificial Intelligence, Machine Learning, Programming, and Computing.
- Supervise undergraduate research projects and mentor students in research methodology and professional practice.
- Develop and review curricula in line with Outcome-Based Education (OBE) and accreditation standards.

- Conduct interdisciplinary research leading to peer-reviewed publications and conference presentations.
- Participate in programme accreditation, examination moderation, quality assurance, and departmental administration.
- Deliver instruction through face-to-face, hybrid, and online learning environments.

Key Achievements

- Supervised **10+ undergraduate engineering projects**.
- Taught classes of **30–50 students**.
- Contributed to curriculum review and programme accreditation.
- Published **10+ peer-reviewed journal articles**.
- Presented research at national and international conferences.
- Integrated Artificial Intelligence and digital technologies into engineering education.

TEACHING & STUDENT SUPERVISION

More than ten years of teaching experience across tertiary, technical, vocational, and secondary education, delivering courses in Engineering, Computing, Information Technology, Mathematics, Physics, Artificial Intelligence, Machine Learning, Programming, and Research Methodology.

Highlights

- Supervised **10+ undergraduate engineering research projects**.
- Mentored students in research design, scientific writing, Artificial Intelligence, programming, and engineering practice.
- Delivered instruction in face-to-face, hybrid, and online learning environments.
- Contributed to curriculum development, course review, and continuous programme improvement.

PREVIOUS TEACHING EXPERIENCE

Mathematics & Technical Drawing Teacher
Port Harcourt International School (2017–2018)
Delivered Mathematics and Technical Drawing instruction, prepared lesson plans, and supported students preparing for external examinations.

Information Technology Instructor
Kufman Power & Automation Training Centre (2015–2016)
Delivered practical Information Technology training and developed learners' digital and workplace computing skills.

Computer Instructor
Bonny Vocational Centre (2012–2013)
Provided vocational ICT training in computer appreciation, office applications, and introductory programming.

Computer & Mathematics Teacher
Kings and Queens Secondary School (2011–2012)

Taught Mathematics and Computer Studies while improving student engagement through learner-centred teaching.

Mathematics & Physics Teacher
St. Paul Comprehensive High School (2010–2011)
Delivered Mathematics and Physics instruction and prepared students for internal and external examinations.

Computer Instructor (NYSC)
Ministry of Information, Ebonyi State (2008–2009)
Conducted ICT training for government personnel and supported digital literacy initiatives.

Industrial Training
Bonny Local Government Area (2005–2006)
Completed industrial attachment focused on engineering operations, technical documentation, and workplace safety.

ACADEMIC LEADERSHIP & ADMINISTRATION

Demonstrated leadership in academic administration, curriculum development, quality assurance, accreditation, and institutional governance.

Head of Department
Department of Electrical/Electronic Engineering Technology

- Led departmental academic and administrative activities.
- Coordinated curriculum review and programme development.
- Supervised academic staff and supported professional development.
- Contributed to accreditation, quality assurance, and institutional planning.

Chairman, Examination Moderation Committee

- Coordinated departmental examination moderation.
- Improved assessment quality assurance and examination integrity.

Chairman, Appraisal Committee

- Led staff performance appraisal and evaluation processes.
- Supported institutional quality improvement and staff development.

Committee Memberships
School Appraisal Committee

- Participated in institutional quality assurance and performance evaluation.

Disciplinary Committee

- Supported institutional governance, policy implementation, and conflict resolution.

RESEARCH & SCHOLARLY CONTRIBUTIONS

Research focuses on Artificial Intelligence, Machine Learning, Computer Vision, Explainable AI, Intelligent Control Systems, and Computational Modelling, with applications in engineering, healthcare, finance, cybersecurity, and public safety.

Research outputs include **10+ peer-reviewed journal publications**, national and international conference presentations, undergraduate research supervision, and interdisciplinary collaborations aimed at addressing real-world engineering challenges.

SELECTED PEER-REVIEWED PUBLICATIONS

2025

Pepple, M.-K. T.

Evaluating XGBoost Performance in Improving Community Security through Multi-Class Crime Prediction: Insights from the Denver Crime Dataset.

International Journal of Applied Sciences and Smart Technologies.

Pepple, M.-K. T.

Dense Neural Networks and Local Interpretable Model-Agnostic Explanations Hybrid Model for Enhanced Interpretability.

International Journal of Computer Science and Mathematical Theory.

Pepple, M.-K. T.

Comparative Analysis of K-Nearest Neighbors, Naïve Bayes and Decision Tree Algorithms for Credit Card Fraud Detection.

World Journal of Innovation and Modern Technology.

Pepple, M.-K. T.

Multicriteria Decision Analysis for Stock Selection: A Comparative Study of ELECTRE III with Veto, TOPSIS and PROMETHEE Methods.

International Journal of Engineering and Modern Technology.

2022

Pepple, M.-K. T.

Comparison of GoogleNet Convolutional Neural Network to Visual Geometry Group Convolutional Neural Network.

Rivers State University Journal of Biology and Applied Sciences.

Pepple, M.-K. T.

Efficiency of State-Dependent Proportional Integral Plus Over Proportional Integral Plus in the Control of Air Conditioner with Free Cooling.

Irish International Journal of Engineering and Scientific Studies.

Pepple, M.-K. T.

QoS Performance Evaluation of Video Streaming Across IPv6.

Rivers State University Journal of Biology and Applied Sciences.

2021

Pepple, M.-K. T.

Energy Efficiency of Model Predictive Control over Proportional Integral-Plus Towards Control.

International Journal of Scientific Research and Engineering Development.

2020

Pepple, M.-K. T.

Impact of Node Velocity on TCP Throughput of Mobile Ad-Hoc Data Network (MADNET).

European Journal of Engineering Technology.

Pepple, M.-K. T.

Experimental Assessment of the Impact of Velocity on UDP End-to-End Delay of Mobile Unicast Data Network (MUDNET).

International Journal of Computing and Technology.

CONFERENCE PRESENTATIONS

International Conferences

- EfficientNet and Swin Transformer Framework with Cross-Attention for Face Recognition, IECEC, University of Nigeria, Nsukka (2026)
- Noise-Aware Feature with Channel Attention U-Net for Clinical Image Denoising, IECEC, University of Nigeria, Nsukka (2026)
- Zero Inflation Count Regression Models, International Conference on Engineering, Management and Statistical Computing (IEMSC), India (2021)
- Markov Switching Regression for Economic Growth, International Conference on Engineering, Management and Statistical Computing (IEMSC), India (2021)

National Conferences

- Fault Detection in Wind Turbine Blades Using Artificial Intelligence Transfer Learning, University of Nigeria, Nsukka (2024)
- Improving Community Safety with XGBoost Machine Learning, Federal Polytechnic of Oil and Gas, Bonny (2024)
- Raw Material Processing and Utilisation for Sustainable Development, Maritime Conference, Federal Polytechnic of Oil and Gas, Bonny (2024)

- Control Systems Optimisation in Air Conditioner Systems, Federal Polytechnic of Oil and Gas, Bonny (2022)
- Shadow Controlled Door Bell Design, Yaba College of Technology (2021)

INSTRUCTIONAL & DIGITAL SKILLS

Higher Education Teaching

- Undergraduate and Technical Education
- Engineering Education
- Curriculum Design and Development
- Outcome-Based Education (OBE)
- Student-Centred Learning
- Project-Based Learning
- Research Supervision
- Academic Assessment and Evaluation
- Examination Moderation
- Quality Assurance in Higher Education
- Academic Mentoring and Advising

Online & Hybrid Teaching

Experienced in designing and delivering engaging learning experiences using contemporary online and hybrid teaching approaches.

Virtual Teaching Platforms

- Zoom
- Microsoft Teams
- Google Meet
- Canvas Learning Management System (LMS)

Digital Learning Competencies

- Online Course Delivery
- Virtual Classroom Facilitation
- Digital Assessment
- Student Engagement Strategies
- Educational Technology Integration
- Collaborative Learning
- Multimedia Instructional Design

TECHNICAL SKILLS

Programming: Python, MATLAB

AI & Machine Learning: Machine Learning, Deep Learning, Computer Vision, Explainable AI (XAI), Predictive Analytics

Frameworks & Libraries: TensorFlow, PyTorch, OpenCV, Scikit-learn

Engineering & Research Tools: MATLAB, Simulink, Jupyter Notebook, Google Colab, VS Code, GitHub, GitHub Copilot, Kaggle

Educational Technologies: Canvas LMS, Zoom, Microsoft Teams, Google Meet

Research Expertise: Data Analysis, Feature Engineering, Model Development, Scientific Computing, Computational Modelling

PROFESSIONAL PROFILES

- Google Scholar
- LinkedIn
- GitHub
- Personal Portfolio

SELECTED ACHIEVEMENTS

- Lecturer I with over 10 years of teaching, research, and academic leadership experience.
- Recipient of the competitive **TETFund Academic Staff Training and Development (AST&D) Scholarship**.
- Author of **10+ peer-reviewed journal publications**.
- Presenter at **9 national and international conferences**.
- Supervised **10+ undergraduate engineering research projects**.
- COREN Registered Engineer with expertise in Artificial Intelligence, Machine Learning, and Computer Vision.
- Experienced in curriculum development, programme accreditation, and academic quality assurance.

REFEREES

Referees available upon request